Software Documentation

1. Software Overview

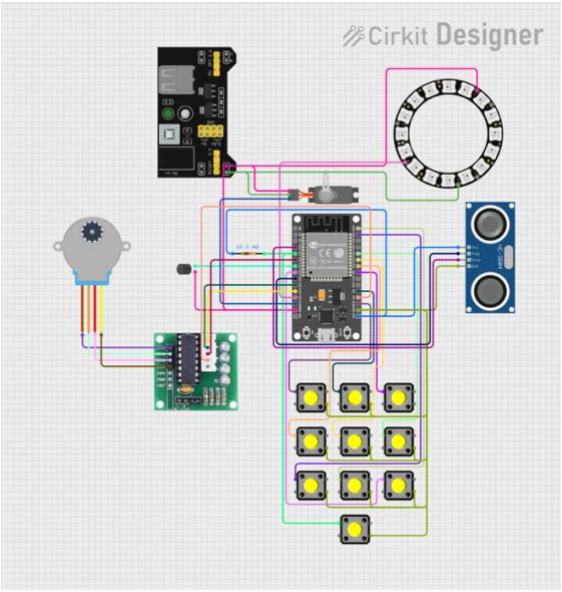
What does the software do?

The software manages a multi-layered security system (three "gateways") for a smart locker. It coordinates communication between a mobile application and an ESP32 microcontroller to verify user identity through various methods:

- **Gateway 1:** A digital One-Time Password (OTP) system requiring physical keypad input.
- **Gateway 2:** A mathematical challenge translated into a physical distance-based gesture using an ultrasonic sensor.
- **Gateway 3:** A color-coded pattern sequence entered via the app to operate a secondary "mini locker."

Platforms Used:

- **MicroPython:** Used for the primary firmware on the ESP32 to handle hardware control (servos, steppers, sensors) and system logic.
- **MIT App Inventor:** Used to create the user interface for generating codes, displaying equations, and sending pattern sequences.
- **Bluetooth Low Energy (BLE):** The communication protocol used to transmit data wirelessly between the smartphone and the locker.
- **Cirkit Designer:** to create a digital circuit of all the components (accurate GPIOs)



2. System Logic

1. **System Initialization:** The ESP32 starts in "Idle Mode," enabling the BLE radio and waiting for a command from the app.

2. **App Command Received:**

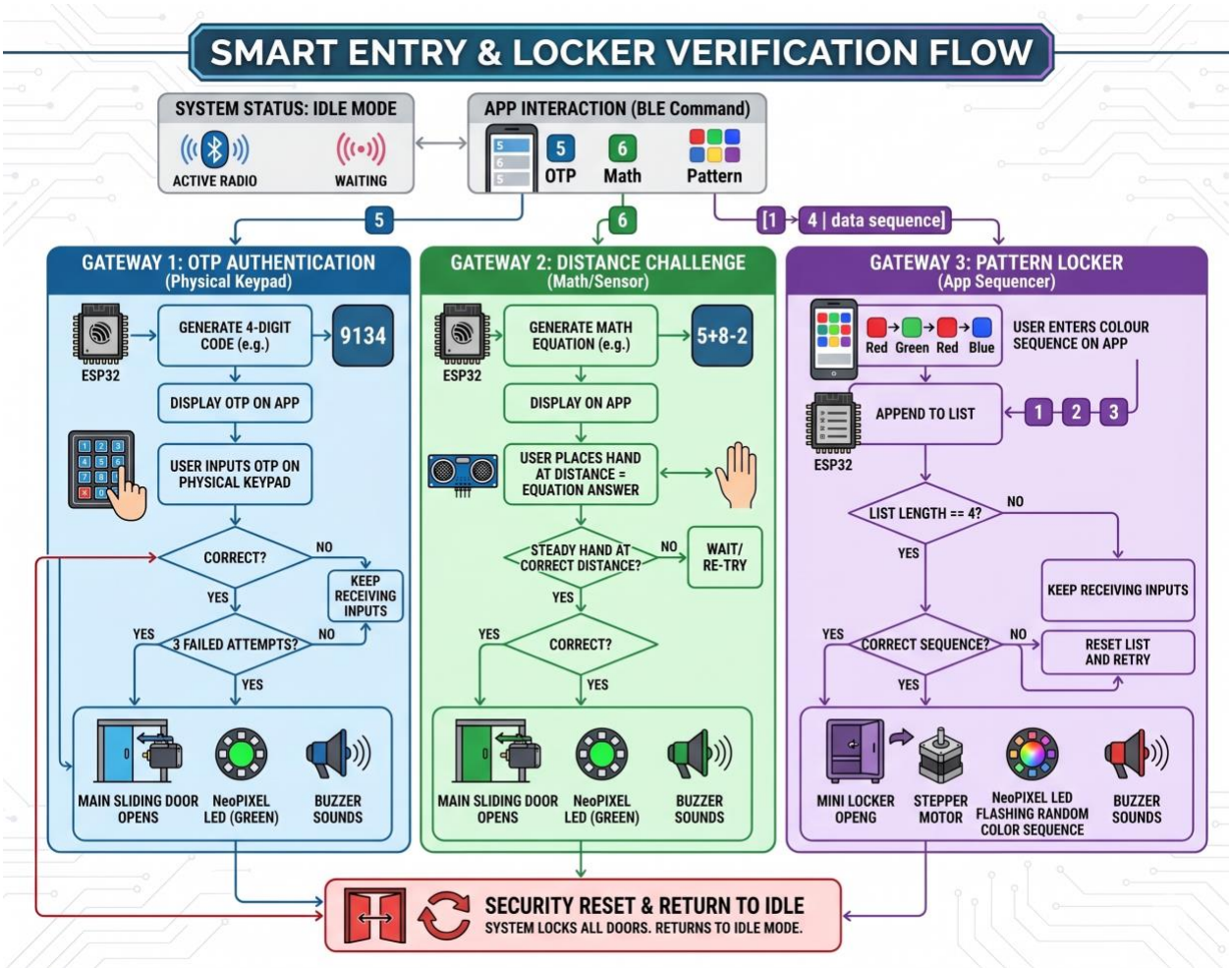
    - **OTP Request ('5'):** App sends '5'; ESP32 generates a 4-digit code, notifies the app, and enters "Gateway 1 Mode".
    - **Math Request ('6'):** App sends '6'; ESP32 generates an equation (e.g., A+B-C), sends it to the app, and enters "Gateway 2 Mode".
    - **Pattern Sequence ('1'-'4'):** User taps colours on the app; app sends the corresponding number.

3. **Physical Verification:**

    - **Gateway 1:** User types the OTP on the physical keypad. If correct, the NeoPixel turns green, the buzzer responds and the main sliding door (Servo) opens.
    - **Gateway 2:** User places their hand at the distance corresponding to the equation's answer. If the sensor detects a steady hand at the correct distance, the door opens, NeoPixel turns green, buzzer responds.
    - **Gateway 3:** If the correct colour sequence is entered on the app, the NeoPixel flashes a random colour sequence and the Mini Locker (Stepper) opens.

4. **Security Reset:** After a successful entry or 3 failed attempts (Gateway 1), the system locks the doors and returns to "Idle Mode"

3. Flowchart



4. App Flow (If MIT App Inventor)

Screen Flow:

- **Connection Screen:** Allows the user to scan for the "Chika" Bluetooth device and connect.
- **Main Interaction Screen:** A dashboard containing buttons for generating the OTP and Math Equation, along with colour buttons for the pattern lock.

Button Actions:

- **"Generate OTP":** Sends text '5' to the ESP32.
- **"Solve to Open":** Sends text '6' to the ESP32.
- **Colour Blocks (Red, Blue, Yellow, etc.):** Send specific digits ('1'-'4') to build the Gateway 3 sequence.

Data Flow:

- **Outgoing:** App writes string commands to the BLE Characteristic UUID.
- **Incoming:** App listens for "Notification" updates from the ESP32 to display the generated OTP or the Math Equation on the screen.